

LiamConnolly_AO3_DataExploration.Rmd

Liam Connolly

September 23, 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
install.packages("tidyverse")
library(tidyverse)
library(lubridate)
library(here)
install.packages('dplyr')
install.packages('ggplot')
```

```
## Warning: package 'ggplot' is not available for this version of R
##
## A version of this package for your version of R might be available elsewhere,
## see the ideas at
## https://cran.r-project.org/doc/manuals/r-patched/R-admin.html#Installing-packages
```

```
install.packages('here')
install.packages("tidyverse")
library(tidyverse)
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

#I was getting a lot of weird errors hence lots of extra code

```
Neonics <- read.csv(file = here("Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"), stringsAsFactors = T)
```

```
Litter <- read.csv(file = here("Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv"), stringsAsFactors = T)
```

#as you can see by the littany of code, I suddenly had no packages

#thankfully I was able to load the datasets eventually, I had a period somewhere that prevented it from

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: There is widespread interest given the relationship insects have with agriculture. They may also impact native biodiversity and other potential ecosystem services. Their presence also has the chance of other negative impacts such as contaminating water supply and/or threatening pollinators.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Litter decomposition is important for nutrient cycling for forests and woody debris plays a role with carbon storage. The woody debris can also serve as an important habitat for many species.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. litter and fine woody debris sampling occurred at terrestrial NEON sites containing woody vegetation >2m tall 2. Target sampling frequency for elevated traps varies by vegetation present at the site, with frequent sampling (1x every 2weeks) in deciduous forest sites during senescence, and in frequent year-round sampling (1x every 1-2 months) at evergreen sites 3. At sites with deciduous vegeta on or limited access during winter months, liter sampling of elevated traps may be discontinued for up to 6 months during the dormant season.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
view(Neonics)
colnames(Neonics)
```

```
## [1] "CAS.Number"           "Chemical.Name"
## [3] "Chemical.Grade"       "Chemical.Analysis.Method"
## [5] "Chemical.Purity"      "Species.Scientific.Name"
## [7] "Species.Common.Name"  "Species.Group"
## [9] "Organism.Lifestage"    "Organism.Age"
## [11] "Organism.Age.Units"    "Exposure.Type"
## [13] "Media.Type"           "Test.Location"
## [15] "Number.of.Doses"       "Conc.1.Type..Author."
## [17] "Conc.1..Author."       "Conc.1.Units..Author."
## [19] "Effect"                "Effect.Measurement"
## [21] "Endpoint"              "Response.Site"
## [23] "Observed.Duration..Days." "Observed.Duration.Units..Days."
## [25] "Author"                 "Reference.Number"
## [27] "Title"                  "Source"
## [29] "Publication.Year"       "Summary.of.Additional.Parameters"
```

```
row.names(Neonics)
```

```
## [1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10"
## [11] "11" "12" "13" "14" "15" "16" "17" "18" "19" "20"
## [21] "21" "22" "23" "24" "25" "26" "27" "28" "29" "30"
## [31] "31" "32" "33" "34" "35" "36" "37" "38" "39" "40"
## [41] "41" "42" "43" "44" "45" "46" "47" "48" "49" "50"
## [51] "51" "52" "53" "54" "55" "56" "57" "58" "59" "60"
## [61] "61" "62" "63" "64" "65" "66" "67" "68" "69" "70"
## [71] "71" "72" "73" "74" "75" "76" "77" "78" "79" "80"
## [81] "81" "82" "83" "84" "85" "86" "87" "88" "89" "90"
## [91] "91" "92" "93" "94" "95" "96" "97" "98" "99" "100"
## [101] "101" "102" "103" "104" "105" "106" "107" "108" "109" "110"
## [111] "111" "112" "113" "114" "115" "116" "117" "118" "119" "120"
## [121] "121" "122" "123" "124" "125" "126" "127" "128" "129" "130"
## [131] "131" "132" "133" "134" "135" "136" "137" "138" "139" "140"
## [141] "141" "142" "143" "144" "145" "146" "147" "148" "149" "150"
## [151] "151" "152" "153" "154" "155" "156" "157" "158" "159" "160"
## [161] "161" "162" "163" "164" "165" "166" "167" "168" "169" "170"
## [171] "171" "172" "173" "174" "175" "176" "177" "178" "179" "180"
## [181] "181" "182" "183" "184" "185" "186" "187" "188" "189" "190"
## [191] "191" "192" "193" "194" "195" "196" "197" "198" "199" "200"
## [201] "201" "202" "203" "204" "205" "206" "207" "208" "209" "210"
## [211] "211" "212" "213" "214" "215" "216" "217" "218" "219" "220"
## [221] "221" "222" "223" "224" "225" "226" "227" "228" "229" "230"
## [231] "231" "232" "233" "234" "235" "236" "237" "238" "239" "240"
## [241] "241" "242" "243" "244" "245" "246" "247" "248" "249" "250"
## [251] "251" "252" "253" "254" "255" "256" "257" "258" "259" "260"
## [261] "261" "262" "263" "264" "265" "266" "267" "268" "269" "270"
## [271] "271" "272" "273" "274" "275" "276" "277" "278" "279" "280"
```

##	[281]	"281"	"282"	"283"	"284"	"285"	"286"	"287"	"288"	"289"	"290"
##	[291]	"291"	"292"	"293"	"294"	"295"	"296"	"297"	"298"	"299"	"300"
##	[301]	"301"	"302"	"303"	"304"	"305"	"306"	"307"	"308"	"309"	"310"
##	[311]	"311"	"312"	"313"	"314"	"315"	"316"	"317"	"318"	"319"	"320"
##	[321]	"321"	"322"	"323"	"324"	"325"	"326"	"327"	"328"	"329"	"330"
##	[331]	"331"	"332"	"333"	"334"	"335"	"336"	"337"	"338"	"339"	"340"
##	[341]	"341"	"342"	"343"	"344"	"345"	"346"	"347"	"348"	"349"	"350"
##	[351]	"351"	"352"	"353"	"354"	"355"	"356"	"357"	"358"	"359"	"360"
##	[361]	"361"	"362"	"363"	"364"	"365"	"366"	"367"	"368"	"369"	"370"
##	[371]	"371"	"372"	"373"	"374"	"375"	"376"	"377"	"378"	"379"	"380"
##	[381]	"381"	"382"	"383"	"384"	"385"	"386"	"387"	"388"	"389"	"390"
##	[391]	"391"	"392"	"393"	"394"	"395"	"396"	"397"	"398"	"399"	"400"
##	[401]	"401"	"402"	"403"	"404"	"405"	"406"	"407"	"408"	"409"	"410"
##	[411]	"411"	"412"	"413"	"414"	"415"	"416"	"417"	"418"	"419"	"420"
##	[421]	"421"	"422"	"423"	"424"	"425"	"426"	"427"	"428"	"429"	"430"
##	[431]	"431"	"432"	"433"	"434"	"435"	"436"	"437"	"438"	"439"	"440"
##	[441]	"441"	"442"	"443"	"444"	"445"	"446"	"447"	"448"	"449"	"450"
##	[451]	"451"	"452"	"453"	"454"	"455"	"456"	"457"	"458"	"459"	"460"
##	[461]	"461"	"462"	"463"	"464"	"465"	"466"	"467"	"468"	"469"	"470"
##	[471]	"471"	"472"	"473"	"474"	"475"	"476"	"477"	"478"	"479"	"480"
##	[481]	"481"	"482"	"483"	"484"	"485"	"486"	"487"	"488"	"489"	"490"
##	[491]	"491"	"492"	"493"	"494"	"495"	"496"	"497"	"498"	"499"	"500"
##	[501]	"501"	"502"	"503"	"504"	"505"	"506"	"507"	"508"	"509"	"510"
##	[511]	"511"	"512"	"513"	"514"	"515"	"516"	"517"	"518"	"519"	"520"
##	[521]	"521"	"522"	"523"	"524"	"525"	"526"	"527"	"528"	"529"	"530"
##	[531]	"531"	"532"	"533"	"534"	"535"	"536"	"537"	"538"	"539"	"540"
##	[541]	"541"	"542"	"543"	"544"	"545"	"546"	"547"	"548"	"549"	"550"
##	[551]	"551"	"552"	"553"	"554"	"555"	"556"	"557"	"558"	"559"	"560"
##	[561]	"561"	"562"	"563"	"564"	"565"	"566"	"567"	"568"	"569"	"570"
##	[571]	"571"	"572"	"573"	"574"	"575"	"576"	"577"	"578"	"579"	"580"
##	[581]	"581"	"582"	"583"	"584"	"585"	"586"	"587"	"588"	"589"	"590"
##	[591]	"591"	"592"	"593"	"594"	"595"	"596"	"597"	"598"	"599"	"600"
##	[601]	"601"	"602"	"603"	"604"	"605"	"606"	"607"	"608"	"609"	"610"
##	[611]	"611"	"612"	"613"	"614"	"615"	"616"	"617"	"618"	"619"	"620"
##	[621]	"621"	"622"	"623"	"624"	"625"	"626"	"627"	"628"	"629"	"630"
##	[631]	"631"	"632"	"633"	"634"	"635"	"636"	"637"	"638"	"639"	"640"
##	[641]	"641"	"642"	"643"	"644"	"645"	"646"	"647"	"648"	"649"	"650"
##	[651]	"651"	"652"	"653"	"654"	"655"	"656"	"657"	"658"	"659"	"660"
##	[661]	"661"	"662"	"663"	"664"	"665"	"666"	"667"	"668"	"669"	"670"
##	[671]	"671"	"672"	"673"	"674"	"675"	"676"	"677"	"678"	"679"	"680"
##	[681]	"681"	"682"	"683"	"684"	"685"	"686"	"687"	"688"	"689"	"690"
##	[691]	"691"	"692"	"693"	"694"	"695"	"696"	"697"	"698"	"699"	"700"
##	[701]	"701"	"702"	"703"	"704"	"705"	"706"	"707"	"708"	"709"	"710"
##	[711]	"711"	"712"	"713"	"714"	"715"	"716"	"717"	"718"	"719"	"720"
##	[721]	"721"	"722"	"723"	"724"	"725"	"726"	"727"	"728"	"729"	"730"
##	[731]	"731"	"732"	"733"	"734"	"735"	"736"	"737"	"738"	"739"	"740"
##	[741]	"741"	"742"	"743"	"744"	"745"	"746"	"747"	"748"	"749"	"750"
##	[751]	"751"	"752"	"753"	"754"	"755"	"756"	"757"	"758"	"759"	"760"
##	[761]	"761"	"762"	"763"	"764"	"765"	"766"	"767"	"768"	"769"	"770"
##	[771]	"771"	"772"	"773"	"774"	"775"	"776"	"777"	"778"	"779"	"780"
##	[781]	"781"	"782"	"783"	"784"	"785"	"786"	"787"	"788"	"789"	"790"
##	[791]	"791"	"792"	"793"	"794"	"795"	"796"	"797"	"798"	"799"	"800"
##	[801]	"801"	"802"	"803"	"804"	"805"	"806"	"807"	"808"	"809"	"810"
##	[811]	"811"	"812"	"813"	"814"	"815"	"816"	"817"	"818"	"819"	"820"

```

## [821] "821" "822" "823" "824" "825" "826" "827" "828" "829" "830"
## [831] "831" "832" "833" "834" "835" "836" "837" "838" "839" "840"
## [841] "841" "842" "843" "844" "845" "846" "847" "848" "849" "850"
## [851] "851" "852" "853" "854" "855" "856" "857" "858" "859" "860"
## [861] "861" "862" "863" "864" "865" "866" "867" "868" "869" "870"
## [871] "871" "872" "873" "874" "875" "876" "877" "878" "879" "880"
## [881] "881" "882" "883" "884" "885" "886" "887" "888" "889" "890"
## [891] "891" "892" "893" "894" "895" "896" "897" "898" "899" "900"
## [901] "901" "902" "903" "904" "905" "906" "907" "908" "909" "910"
## [911] "911" "912" "913" "914" "915" "916" "917" "918" "919" "920"
## [921] "921" "922" "923" "924" "925" "926" "927" "928" "929" "930"
## [931] "931" "932" "933" "934" "935" "936" "937" "938" "939" "940"
## [941] "941" "942" "943" "944" "945" "946" "947" "948" "949" "950"
## [951] "951" "952" "953" "954" "955" "956" "957" "958" "959" "960"
## [961] "961" "962" "963" "964" "965" "966" "967" "968" "969" "970"
## [971] "971" "972" "973" "974" "975" "976" "977" "978" "979" "980"
## [981] "981" "982" "983" "984" "985" "986" "987" "988" "989" "990"
## [991] "991" "992" "993" "994" "995" "996" "997" "998" "999" "1000"
## [1001] "1001" "1002" "1003" "1004" "1005" "1006" "1007" "1008" "1009" "1010"
## [1011] "1011" "1012" "1013" "1014" "1015" "1016" "1017" "1018" "1019" "1020"
## [1021] "1021" "1022" "1023" "1024" "1025" "1026" "1027" "1028" "1029" "1030"
## [1031] "1031" "1032" "1033" "1034" "1035" "1036" "1037" "1038" "1039" "1040"
## [1041] "1041" "1042" "1043" "1044" "1045" "1046" "1047" "1048" "1049" "1050"
## [1051] "1051" "1052" "1053" "1054" "1055" "1056" "1057" "1058" "1059" "1060"
## [1061] "1061" "1062" "1063" "1064" "1065" "1066" "1067" "1068" "1069" "1070"
## [1071] "1071" "1072" "1073" "1074" "1075" "1076" "1077" "1078" "1079" "1080"
## [1081] "1081" "1082" "1083" "1084" "1085" "1086" "1087" "1088" "1089" "1090"
## [1091] "1091" "1092" "1093" "1094" "1095" "1096" "1097" "1098" "1099" "1100"
## [1101] "1101" "1102" "1103" "1104" "1105" "1106" "1107" "1108" "1109" "1110"
## [1111] "1111" "1112" "1113" "1114" "1115" "1116" "1117" "1118" "1119" "1120"
## [1121] "1121" "1122" "1123" "1124" "1125" "1126" "1127" "1128" "1129" "1130"
## [1131] "1131" "1132" "1133" "1134" "1135" "1136" "1137" "1138" "1139" "1140"
## [1141] "1141" "1142" "1143" "1144" "1145" "1146" "1147" "1148" "1149" "1150"
## [1151] "1151" "1152" "1153" "1154" "1155" "1156" "1157" "1158" "1159" "1160"
## [1161] "1161" "1162" "1163" "1164" "1165" "1166" "1167" "1168" "1169" "1170"
## [1171] "1171" "1172" "1173" "1174" "1175" "1176" "1177" "1178" "1179" "1180"
## [1181] "1181" "1182" "1183" "1184" "1185" "1186" "1187" "1188" "1189" "1190"
## [1191] "1191" "1192" "1193" "1194" "1195" "1196" "1197" "1198" "1199" "1200"
## [1201] "1201" "1202" "1203" "1204" "1205" "1206" "1207" "1208" "1209" "1210"
## [1211] "1211" "1212" "1213" "1214" "1215" "1216" "1217" "1218" "1219" "1220"
## [1221] "1221" "1222" "1223" "1224" "1225" "1226" "1227" "1228" "1229" "1230"
## [1231] "1231" "1232" "1233" "1234" "1235" "1236" "1237" "1238" "1239" "1240"
## [1241] "1241" "1242" "1243" "1244" "1245" "1246" "1247" "1248" "1249" "1250"
## [1251] "1251" "1252" "1253" "1254" "1255" "1256" "1257" "1258" "1259" "1260"
## [1261] "1261" "1262" "1263" "1264" "1265" "1266" "1267" "1268" "1269" "1270"
## [1271] "1271" "1272" "1273" "1274" "1275" "1276" "1277" "1278" "1279" "1280"
## [1281] "1281" "1282" "1283" "1284" "1285" "1286" "1287" "1288" "1289" "1290"
## [1291] "1291" "1292" "1293" "1294" "1295" "1296" "1297" "1298" "1299" "1300"
## [1301] "1301" "1302" "1303" "1304" "1305" "1306" "1307" "1308" "1309" "1310"
## [1311] "1311" "1312" "1313" "1314" "1315" "1316" "1317" "1318" "1319" "1320"
## [1321] "1321" "1322" "1323" "1324" "1325" "1326" "1327" "1328" "1329" "1330"
## [1331] "1331" "1332" "1333" "1334" "1335" "1336" "1337" "1338" "1339" "1340"
## [1341] "1341" "1342" "1343" "1344" "1345" "1346" "1347" "1348" "1349" "1350"
## [1351] "1351" "1352" "1353" "1354" "1355" "1356" "1357" "1358" "1359" "1360"

```

##	[1361]	"1361"	"1362"	"1363"	"1364"	"1365"	"1366"	"1367"	"1368"	"1369"	"1370"
##	[1371]	"1371"	"1372"	"1373"	"1374"	"1375"	"1376"	"1377"	"1378"	"1379"	"1380"
##	[1381]	"1381"	"1382"	"1383"	"1384"	"1385"	"1386"	"1387"	"1388"	"1389"	"1390"
##	[1391]	"1391"	"1392"	"1393"	"1394"	"1395"	"1396"	"1397"	"1398"	"1399"	"1400"
##	[1401]	"1401"	"1402"	"1403"	"1404"	"1405"	"1406"	"1407"	"1408"	"1409"	"1410"
##	[1411]	"1411"	"1412"	"1413"	"1414"	"1415"	"1416"	"1417"	"1418"	"1419"	"1420"
##	[1421]	"1421"	"1422"	"1423"	"1424"	"1425"	"1426"	"1427"	"1428"	"1429"	"1430"
##	[1431]	"1431"	"1432"	"1433"	"1434"	"1435"	"1436"	"1437"	"1438"	"1439"	"1440"
##	[1441]	"1441"	"1442"	"1443"	"1444"	"1445"	"1446"	"1447"	"1448"	"1449"	"1450"
##	[1451]	"1451"	"1452"	"1453"	"1454"	"1455"	"1456"	"1457"	"1458"	"1459"	"1460"
##	[1461]	"1461"	"1462"	"1463"	"1464"	"1465"	"1466"	"1467"	"1468"	"1469"	"1470"
##	[1471]	"1471"	"1472"	"1473"	"1474"	"1475"	"1476"	"1477"	"1478"	"1479"	"1480"
##	[1481]	"1481"	"1482"	"1483"	"1484"	"1485"	"1486"	"1487"	"1488"	"1489"	"1490"
##	[1491]	"1491"	"1492"	"1493"	"1494"	"1495"	"1496"	"1497"	"1498"	"1499"	"1500"
##	[1501]	"1501"	"1502"	"1503"	"1504"	"1505"	"1506"	"1507"	"1508"	"1509"	"1510"
##	[1511]	"1511"	"1512"	"1513"	"1514"	"1515"	"1516"	"1517"	"1518"	"1519"	"1520"
##	[1521]	"1521"	"1522"	"1523"	"1524"	"1525"	"1526"	"1527"	"1528"	"1529"	"1530"
##	[1531]	"1531"	"1532"	"1533"	"1534"	"1535"	"1536"	"1537"	"1538"	"1539"	"1540"
##	[1541]	"1541"	"1542"	"1543"	"1544"	"1545"	"1546"	"1547"	"1548"	"1549"	"1550"
##	[1551]	"1551"	"1552"	"1553"	"1554"	"1555"	"1556"	"1557"	"1558"	"1559"	"1560"
##	[1561]	"1561"	"1562"	"1563"	"1564"	"1565"	"1566"	"1567"	"1568"	"1569"	"1570"
##	[1571]	"1571"	"1572"	"1573"	"1574"	"1575"	"1576"	"1577"	"1578"	"1579"	"1580"
##	[1581]	"1581"	"1582"	"1583"	"1584"	"1585"	"1586"	"1587"	"1588"	"1589"	"1590"
##	[1591]	"1591"	"1592"	"1593"	"1594"	"1595"	"1596"	"1597"	"1598"	"1599"	"1600"
##	[1601]	"1601"	"1602"	"1603"	"1604"	"1605"	"1606"	"1607"	"1608"	"1609"	"1610"
##	[1611]	"1611"	"1612"	"1613"	"1614"	"1615"	"1616"	"1617"	"1618"	"1619"	"1620"
##	[1621]	"1621"	"1622"	"1623"	"1624"	"1625"	"1626"	"1627"	"1628"	"1629"	"1630"
##	[1631]	"1631"	"1632"	"1633"	"1634"	"1635"	"1636"	"1637"	"1638"	"1639"	"1640"
##	[1641]	"1641"	"1642"	"1643"	"1644"	"1645"	"1646"	"1647"	"1648"	"1649"	"1650"
##	[1651]	"1651"	"1652"	"1653"	"1654"	"1655"	"1656"	"1657"	"1658"	"1659"	"1660"
##	[1661]	"1661"	"1662"	"1663"	"1664"	"1665"	"1666"	"1667"	"1668"	"1669"	"1670"
##	[1671]	"1671"	"1672"	"1673"	"1674"	"1675"	"1676"	"1677"	"1678"	"1679"	"1680"
##	[1681]	"1681"	"1682"	"1683"	"1684"	"1685"	"1686"	"1687"	"1688"	"1689"	"1690"
##	[1691]	"1691"	"1692"	"1693"	"1694"	"1695"	"1696"	"1697"	"1698"	"1699"	"1700"
##	[1701]	"1701"	"1702"	"1703"	"1704"	"1705"	"1706"	"1707"	"1708"	"1709"	"1710"
##	[1711]	"1711"	"1712"	"1713"	"1714"	"1715"	"1716"	"1717"	"1718"	"1719"	"1720"
##	[1721]	"1721"	"1722"	"1723"	"1724"	"1725"	"1726"	"1727"	"1728"	"1729"	"1730"
##	[1731]	"1731"	"1732"	"1733"	"1734"	"1735"					

```

## [1901] "1901" "1902" "1903" "1904" "1905" "1906" "1907" "1908" "1909" "1910"
## [1911] "1911" "1912" "1913" "1914" "1915" "1916" "1917" "1918" "1919" "1920"
## [1921] "1921" "1922" "1923" "1924" "1925" "1926" "1927" "1928" "1929" "1930"
## [1931] "1931" "1932" "1933" "1934" "1935" "1936" "1937" "1938" "1939" "1940"
## [1941] "1941" "1942" "1943" "1944" "1945" "1946" "1947" "1948" "1949" "1950"
## [1951] "1951" "1952" "1953" "1954" "1955" "1956" "1957" "1958" "1959" "1960"
## [1961] "1961" "1962" "1963" "1964" "1965" "1966" "1967" "1968" "1969" "1970"
## [1971] "1971" "1972" "1973" "1974" "1975" "1976" "1977" "1978" "1979" "1980"
## [1981] "1981" "1982" "1983" "1984" "1985" "1986" "1987" "1988" "1989" "1990"
## [1991] "1991" "1992" "1993" "1994" "1995" "1996" "1997" "1998" "1999" "2000"
## [2001] "2001" "2002" "2003" "2004" "2005" "2006" "2007" "2008" "2009" "2010"
## [2011] "2011" "2012" "2013" "2014" "2015" "2016" "2017" "2018" "2019" "2020"
## [2021] "2021" "2022" "2023" "2024" "2025" "2026" "2027" "2028" "2029" "2030"
## [2031] "2031" "2032" "2033" "2034" "2035" "2036" "2037" "2038" "2039" "2040"
## [2041] "2041" "2042" "2043" "2044" "2045" "2046" "2047" "2048" "2049" "2050"
## [2051] "2051" "2052" "2053" "2054" "2055" "2056" "2057" "2058" "2059" "2060"
## [2061] "2061" "2062" "2063" "2064" "2065" "2066" "2067" "2068" "2069" "2070"
## [2071] "2071" "2072" "2073" "2074" "2075" "2076" "2077" "2078" "2079" "2080"
## [2081] "2081" "2082" "2083" "2084" "2085" "2086" "2087" "2088" "2089" "2090"
## [2091] "2091" "2092" "2093" "2094" "2095" "2096" "2097" "2098" "2099" "2100"
## [2101] "2101" "2102" "2103" "2104" "2105" "2106" "2107" "2108" "2109" "2110"
## [2111] "2111" "2112" "2113" "2114" "2115" "2116" "2117" "2118" "2119" "2120"
## [2121] "2121" "2122" "2123" "2124" "2125" "2126" "2127" "2128" "2129" "2130"
## [2131] "2131" "2132" "2133" "2134" "2135" "2136" "2137" "2138" "2139" "2140"
## [2141] "2141" "2142" "2143" "2144" "2145" "2146" "2147" "2148" "2149" "2150"
## [2151] "2151" "2152" "2153" "2154" "2155" "2156" "2157" "2158" "2159" "2160"
## [2161] "2161" "2162" "2163" "2164" "2165" "2166" "2167" "2168" "2169" "2170"
## [2171] "2171" "2172" "2173" "2174" "2175" "2176" "2177" "2178" "2179" "2180"
## [2181] "2181" "2182" "2183" "2184" "2185" "2186" "2187" "2188" "2189" "2190"
## [2191] "2191" "2192" "2193" "2194" "2195" "2196" "2197" "2198" "2199" "2200"
## [2201] "2201" "2202" "2203" "2204" "2205" "2206" "2207" "2208" "2209" "2210"
## [2211] "2211" "2212" "2213" "2214" "2215" "2216" "2217" "2218" "2219" "2220"
## [2221] "2221" "2222" "2223" "2224" "2225" "2226" "2227" "2228" "2229" "2230"
## [2231] "2231" "2232" "2233" "2234" "2235" "2236" "2237" "2238" "2239" "2240"
## [2241] "2241" "2242" "2243" "2244" "2245" "2246" "2247" "2248" "2249" "2250"
## [2251] "2251" "2252" "2253" "2254" "2255" "2256" "2257" "2258" "2259" "2260"
## [2261] "2261" "2262" "2263" "2264" "2265" "2266" "2267" "2268" "2269" "2270"
## [2271] "2271" "2272" "2273" "2274" "2275" "2276" "2277" "2278" "2279" "2280"
## [2281] "2281" "2282" "2283" "2284" "2285" "2286" "2287" "2288" "2289" "2290"
## [2291] "2291" "2292" "2293" "2294" "2295" "2296" "2297" "2298" "2299" "2300"
## [2301] "2301" "2302" "2303" "2304" "2305" "2306" "2307" "2308" "2309" "2310"
## [2311] "2311" "2312" "2313" "2314" "2315" "2316" "2317" "2318" "2319" "2320"
## [2321] "2321" "2322" "2323" "2324" "2325" "2326" "2327" "2328" "2329" "2330"
## [2331] "2331" "2332" "2333" "2334" "2335" "2336" "2337" "2338" "2339" "2340"
## [2341] "2341" "2342" "2343" "2344" "2345" "2346" "2347" "2348" "2349" "2350"
## [2351] "2351" "2352" "2353" "2354" "2355" "2356" "2357" "2358" "2359" "2360"
## [2361] "2361" "2362" "2363" "2364" "2365" "2366" "2367" "2368" "2369" "2370"
## [2371] "2371" "2372" "2373" "2374" "2375" "2376" "2377" "2378" "2379" "2380"
## [2381] "2381" "2382" "2383" "2384" "2385" "2386" "2387" "2388" "2389" "2390"
## [2391] "2391" "2392" "2393" "2394" "2395" "2396" "2397" "2398" "2399" "2400"
## [2401] "2401" "2402" "2403" "2404" "2405" "2406" "2407" "2408" "2409" "2410"
## [2411] "2411" "2412" "2413" "2414" "2415" "2416" "2417" "2418" "2419" "2420"
## [2421] "2421" "2422" "2423" "2424" "2425" "2426" "2427" "2428" "2429" "2430"
## [2431] "2431" "2432" "2433" "2434" "2435" "2436" "2437" "2438" "2439" "2440"

```

##	[2441]	"2441"	"2442"	"2443"	"2444"	"2445"	"2446"	"2447"	"2448"	"2449"	"2450"
##	[2451]	"2451"	"2452"	"2453"	"2454"	"2455"	"2456"	"2457"	"2458"	"2459"	"2460"
##	[2461]	"2461"	"2462"	"2463"	"2464"	"2465"	"2466"	"2467"	"2468"	"2469"	"2470"
##	[2471]	"2471"	"2472"	"2473"	"2474"	"2475"	"2476"	"2477"	"2478"	"2479"	"2480"
##	[2481]	"2481"	"2482"	"2483"	"2484"	"2485"	"2486"	"2487"	"2488"	"2489"	"2490"
##	[2491]	"2491"	"2492"	"2493"	"2494"	"2495"	"2496"	"2497"	"2498"	"2499"	"2500"
##	[2501]	"2501"	"2502"	"2503"	"2504"	"2505"	"2506"	"2507"	"2508"	"2509"	"2510"
##	[2511]	"2511"	"2512"	"2513"	"2514"	"2515"	"2516"	"2517"	"2518"	"2519"	"2520"
##	[2521]	"2521"	"2522"	"2523"	"2524"	"2525"	"2526"	"2527"	"2528"	"2529"	"2530"
##	[2531]	"2531"	"2532"	"2533"	"2534"	"2535"	"2536"	"2537"	"2538"	"2539"	"2540"
##	[2541]	"2541"	"2542"	"2543"	"2544"	"2545"	"2546"	"2547"	"2548"	"2549"	"2550"
##	[2551]	"2551"	"2552"	"2553"	"2554"	"2555"	"2556"	"2557"	"2558"	"2559"	"2560"
##	[2561]	"2561"	"2562"	"2563"	"2564"	"2565"	"2566"	"2567"	"2568"	"2569"	"2570"
##	[2571]	"2571"	"2572"	"2573"	"2574"	"2575"	"2576"	"2577"	"2578"	"2579"	"2580"
##	[2581]	"2581"	"2582"	"2583"	"2584"	"2585"	"2586"	"2587"	"2588"	"2589"	"2590"
##	[2591]	"2591"	"2592"	"2593"	"2594"	"2595"	"2596"	"2597"	"2598"	"2599"	"2600"
##	[2601]	"2601"	"2602"	"2603"	"2604"	"2605"	"2606"	"2607"	"2608"	"2609"	"2610"
##	[2611]	"2611"	"2612"	"2613"	"2614"	"2615"	"2616"	"2617"	"2618"	"2619"	"2620"
##	[2621]	"2621"	"2622"	"2623"	"2624"	"2625"	"2626"	"2627"	"2628"	"2629"	"2630"
##	[2631]	"2631"	"2632"	"2633"	"2634"	"2635"	"2636"	"2637"	"2638"	"2639"	"2640"
##	[2641]	"2641"	"2642"	"2643"	"2644"	"2645"	"2646"	"2647"	"2648"	"2649"	"2650"
##	[2651]	"2651"	"2652"	"2653"	"2654"	"2655"	"2656"	"2657"	"2658"	"2659"	"2660"
##	[2661]	"2661"	"2662"	"2663"	"2664"	"2665"	"2666"	"2667"	"2668"	"2669"	"2670"
##	[2671]	"2671"	"2672"	"2673"	"2674"	"2675"	"2676"	"2677"	"2678"	"2679"	"2680"
##	[2681]	"2681"	"2682"	"2683"	"2684"	"2685"	"2686"	"2687"	"2688"	"2689"	"2690"
##	[2691]	"2691"	"2692"	"2693"	"2694"	"2695"	"2696"	"2697"	"2698"	"2699"	"2700"
##	[2701]	"2701"	"2702"	"2703"	"2704"	"2705"	"2706"	"2707"	"2708"	"2709"	"2710"
##	[2711]	"2711"	"2712"	"2713"	"2714"	"2715"	"2716"	"2717"	"2718"	"2719"	"2720"
##	[2721]	"2721"	"2722"	"2723"	"2724"	"2725"	"2726"	"2727"	"2728"	"2729"	"2730"
##	[2731]	"2731"	"2732"	"2733"	"2734"	"2735"	"2736"	"2737"	"2738"	"2739"	"2740"
##	[2741]	"2741"	"2742"	"2743"	"2744"	"2745"	"2746"	"2747"	"2748"	"2749"	"2750"
##	[2751]	"2751"	"2752"	"2753"	"2754"	"2755"	"2756"	"2757"	"2758"	"2759"	"2760"
##	[2761]	"2761"	"2762"	"2763"	"2764"	"2765"	"2766"	"2767"	"2768"	"2769"	"2770"
##	[2771]	"2771"	"2772"	"2773"	"2774"	"2775"	"2776"	"2777"	"2778"	"2779"	"2780"
##	[2781]	"2781"	"2782"	"2783"	"2784"	"2785"	"2786"	"2787"	"2788"	"2789"	"2790"
##	[2791]	"2791"	"2792"	"2793"	"2794"	"2795"	"2796"	"2797"	"2798"	"2799"	"2800"
##	[2801]	"2801"	"2802"	"2803"	"2804"	"2805"	"2806"	"2807"	"2808"	"2809"	"2810"
##	[2811]	"2811"	"2812"	"2813"	"2814"	"2815"					

```

## [2981] "2981" "2982" "2983" "2984" "2985" "2986" "2987" "2988" "2989" "2990"
## [2991] "2991" "2992" "2993" "2994" "2995" "2996" "2997" "2998" "2999" "3000"
## [3001] "3001" "3002" "3003" "3004" "3005" "3006" "3007" "3008" "3009" "3010"
## [3011] "3011" "3012" "3013" "3014" "3015" "3016" "3017" "3018" "3019" "3020"
## [3021] "3021" "3022" "3023" "3024" "3025" "3026" "3027" "3028" "3029" "3030"
## [3031] "3031" "3032" "3033" "3034" "3035" "3036" "3037" "3038" "3039" "3040"
## [3041] "3041" "3042" "3043" "3044" "3045" "3046" "3047" "3048" "3049" "3050"
## [3051] "3051" "3052" "3053" "3054" "3055" "3056" "3057" "3058" "3059" "3060"
## [3061] "3061" "3062" "3063" "3064" "3065" "3066" "3067" "3068" "3069" "3070"
## [3071] "3071" "3072" "3073" "3074" "3075" "3076" "3077" "3078" "3079" "3080"
## [3081] "3081" "3082" "3083" "3084" "3085" "3086" "3087" "3088" "3089" "3090"
## [3091] "3091" "3092" "3093" "3094" "3095" "3096" "3097" "3098" "3099" "3100"
## [3101] "3101" "3102" "3103" "3104" "3105" "3106" "3107" "3108" "3109" "3110"
## [3111] "3111" "3112" "3113" "3114" "3115" "3116" "3117" "3118" "3119" "3120"
## [3121] "3121" "3122" "3123" "3124" "3125" "3126" "3127" "3128" "3129" "3130"
## [3131] "3131" "3132" "3133" "3134" "3135" "3136" "3137" "3138" "3139" "3140"
## [3141] "3141" "3142" "3143" "3144" "3145" "3146" "3147" "3148" "3149" "3150"
## [3151] "3151" "3152" "3153" "3154" "3155" "3156" "3157" "3158" "3159" "3160"
## [3161] "3161" "3162" "3163" "3164" "3165" "3166" "3167" "3168" "3169" "3170"
## [3171] "3171" "3172" "3173" "3174" "3175" "3176" "3177" "3178" "3179" "3180"
## [3181] "3181" "3182" "3183" "3184" "3185" "3186" "3187" "3188" "3189" "3190"
## [3191] "3191" "3192" "3193" "3194" "3195" "3196" "3197" "3198" "3199" "3200"
## [3201] "3201" "3202" "3203" "3204" "3205" "3206" "3207" "3208" "3209" "3210"
## [3211] "3211" "3212" "3213" "3214" "3215" "3216" "3217" "3218" "3219" "3220"
## [3221] "3221" "3222" "3223" "3224" "3225" "3226" "3227" "3228" "3229" "3230"
## [3231] "3231" "3232" "3233" "3234" "3235" "3236" "3237" "3238" "3239" "3240"
## [3241] "3241" "3242" "3243" "3244" "3245" "3246" "3247" "3248" "3249" "3250"
## [3251] "3251" "3252" "3253" "3254" "3255" "3256" "3257" "3258" "3259" "3260"
## [3261] "3261" "3262" "3263" "3264" "3265" "3266" "3267" "3268" "3269" "3270"
## [3271] "3271" "3272" "3273" "3274" "3275" "3276" "3277" "3278" "3279" "3280"
## [3281] "3281" "3282" "3283" "3284" "3285" "3286" "3287" "3288" "3289" "3290"
## [3291] "3291" "3292" "3293" "3294" "3295" "3296" "3297" "3298" "3299" "3300"
## [3301] "3301" "3302" "3303" "3304" "3305" "3306" "3307" "3308" "3309" "3310"
## [3311] "3311" "3312" "3313" "3314" "3315" "3316" "3317" "3318" "3319" "3320"
## [3321] "3321" "3322" "3323" "3324" "3325" "3326" "3327" "3328" "3329" "3330"
## [3331] "3331" "3332" "3333" "3334" "3335" "3336" "3337" "3338" "3339" "3340"
## [3341] "3341" "3342" "3343" "3344" "3345" "3346" "3347" "3348" "3349" "3350"
## [3351] "3351" "3352" "3353" "3354" "3355" "3356" "3357" "3358" "3359" "3360"
## [3361] "3361" "3362" "3363" "3364" "3365" "3366" "3367" "3368" "3369" "3370"
## [3371] "3371" "3372" "3373" "3374" "3375" "3376" "3377" "3378" "3379" "3380"
## [3381] "3381" "3382" "3383" "3384" "3385" "3386" "3387" "3388" "3389" "3390"
## [3391] "3391" "3392" "3393" "3394" "3395" "3396" "3397" "3398" "3399" "3400"
## [3401] "3401" "3402" "3403" "3404" "3405" "3406" "3407" "3408" "3409" "3410"
## [3411] "3411" "3412" "3413" "3414" "3415" "3416" "3417" "3418" "3419" "3420"
## [3421] "3421" "3422" "3423" "3424" "3425" "3426" "3427" "3428" "3429" "3430"
## [3431] "3431" "3432" "3433" "3434" "3435" "3436" "3437" "3438" "3439" "3440"
## [3441] "3441" "3442" "3443" "3444" "3445" "3446" "3447" "3448" "3449" "3450"
## [3451] "3451" "3452" "3453" "3454" "3455" "3456" "3457" "3458" "3459" "3460"
## [3461] "3461" "3462" "3463" "3464" "3465" "3466" "3467" "3468" "3469" "3470"
## [3471] "3471" "3472" "3473" "3474" "3475" "3476" "3477" "3478" "3479" "3480"
## [3481] "3481" "3482" "3483" "3484" "3485" "3486" "3487" "3488" "3489" "3490"
## [3491] "3491" "3492" "3493" "3494" "3495" "3496" "3497" "3498" "3499" "3500"
## [3501] "3501" "3502" "3503" "3504" "3505" "3506" "3507" "3508" "3509" "3510"
## [3511] "3511" "3512" "3513" "3514" "3515" "3516" "3517" "3518" "3519" "3520"

```

##	[3521]	"3521"	"3522"	"3523"	"3524"	"3525"	"3526"	"3527"	"3528"	"3529"	"3530"
##	[3531]	"3531"	"3532"	"3533"	"3534"	"3535"	"3536"	"3537"	"3538"	"3539"	"3540"
##	[3541]	"3541"	"3542"	"3543"	"3544"	"3545"	"3546"	"3547"	"3548"	"3549"	"3550"
##	[3551]	"3551"	"3552"	"3553"	"3554"	"3555"	"3556"	"3557"	"3558"	"3559"	"3560"
##	[3561]	"3561"	"3562"	"3563"	"3564"	"3565"	"3566"	"3567"	"3568"	"3569"	"3570"
##	[3571]	"3571"	"3572"	"3573"	"3574"	"3575"	"3576"	"3577"	"3578"	"3579"	"3580"
##	[3581]	"3581"	"3582"	"3583"	"3584"	"3585"	"3586"	"3587"	"3588"	"3589"	"3590"
##	[3591]	"3591"	"3592"	"3593"	"3594"	"3595"	"3596"	"3597"	"3598"	"3599"	"3600"
##	[3601]	"3601"	"3602"	"3603"	"3604"	"3605"	"3606"	"3607"	"3608"	"3609"	"3610"
##	[3611]	"3611"	"3612"	"3613"	"3614"	"3615"	"3616"	"3617"	"3618"	"3619"	"3620"
##	[3621]	"3621"	"3622"	"3623"	"3624"	"3625"	"3626"	"3627"	"3628"	"3629"	"3630"
##	[3631]	"3631"	"3632"	"3633"	"3634"	"3635"	"3636"	"3637"	"3638"	"3639"	"3640"
##	[3641]	"3641"	"3642"	"3643"	"3644"	"3645"	"3646"	"3647"	"3648"	"3649"	"3650"
##	[3651]	"3651"	"3652"	"3653"	"3654"	"3655"	"3656"	"3657"	"3658"	"3659"	"3660"
##	[3661]	"3661"	"3662"	"3663"	"3664"	"3665"	"3666"	"3667"	"3668"	"3669"	"3670"
##	[3671]	"3671"	"3672"	"3673"	"3674"	"3675"	"3676"	"3677"	"3678"	"3679"	"3680"
##	[3681]	"3681"	"3682"	"3683"	"3684"	"3685"	"3686"	"3687"	"3688"	"3689"	"3690"
##	[3691]	"3691"	"3692"	"3693"	"3694"	"3695"	"3696"	"3697"	"3698"	"3699"	"3700"
##	[3701]	"3701"	"3702"	"3703"	"3704"	"3705"	"3706"	"3707"	"3708"	"3709"	"3710"
##	[3711]	"3711"	"3712"	"3713"	"3714"	"3715"	"3716"	"3717"	"3718"	"3719"	"3720"
##	[3721]	"3721"	"3722"	"3723"	"3724"	"3725"	"3726"	"3727"	"3728"	"3729"	"3730"
##	[3731]	"3731"	"3732"	"3733"	"3734"	"3735"	"3736"	"3737"	"3738"	"3739"	"3740"
##	[3741]	"3741"	"3742"	"3743"	"3744"	"3745"	"3746"	"3747"	"3748"	"3749"	"3750"
##	[3751]	"3751"	"3752"	"3753"	"3754"	"3755"	"3756"	"3757"	"3758"	"3759"	"3760"
##	[3761]	"3761"	"3762"	"3763"	"3764"	"3765"	"3766"	"3767"	"3768"	"3769"	"3770"
##	[3771]	"3771"	"3772"	"3773"	"3774"	"3775"	"3776"	"3777"	"3778"	"3779"	"3780"
##	[3781]	"3781"	"3782"	"3783"	"3784"	"3785"	"3786"	"3787"	"3788"	"3789"	"3790"
##	[3791]	"3791"	"3792"	"3793"	"3794"	"3795"	"3796"	"3797"	"3798"	"3799"	"3800"
##	[3801]	"3801"	"3802"	"3803"	"3804"	"3805"	"3806"	"3807"	"3808"	"3809"	"3810"
##	[3811]	"3811"	"3812"	"3813"	"3814"	"3815"	"3816"	"3817"	"3818"	"3819"	"3820"
##	[3821]	"3821"	"3822"	"3823"	"3824"	"3825"	"3826"	"3827"	"3828"	"3829"	"3830"
##	[3831]	"3831"	"3832"	"3833"	"3834"	"3835"	"3836"	"3837"	"3838"	"3839"	"3840"
##	[3841]	"3841"	"3842"	"3843"	"3844"	"3845"	"3846"	"3847"	"3848"	"3849"	"3850"
##	[3851]	"3851"	"3852"	"3853"	"3854"	"3855"	"3856"	"3857"	"3858"	"3859"	"3860"
##	[3861]	"3861"	"3862"	"3863"	"3864"	"3865"	"3866"	"3867"	"3868"	"3869"	"3870"
##	[3871]	"3871"	"3872"	"3873"	"3874"	"3875"	"3876"	"3877"	"3878"	"3879"	"3880"
##	[3881]	"3881"	"3882"	"3883"	"3884"	"3885"	"3886"	"3887"	"3888"	"3889"	"3890"
##	[3891]	"3891"	"3892"	"3893"	"3894"	"3895"					

##	[4061]	"4061"	"4062"	"4063"	"4064"	"4065"	"4066"	"4067"	"4068"	"4069"	"4070"
##	[4071]	"4071"	"4072"	"4073"	"4074"	"4075"	"4076"	"4077"	"4078"	"4079"	"4080"
##	[4081]	"4081"	"4082"	"4083"	"4084"	"4085"	"4086"	"4087"	"4088"	"4089"	"4090"
##	[4091]	"4091"	"4092"	"4093"	"4094"	"4095"	"4096"	"4097"	"4098"	"4099"	"4100"
##	[4101]	"4101"	"4102"	"4103"	"4104"	"4105"	"4106"	"4107"	"4108"	"4109"	"4110"
##	[4111]	"4111"	"4112"	"4113"	"4114"	"4115"	"4116"	"4117"	"4118"	"4119"	"4120"
##	[4121]	"4121"	"4122"	"4123"	"4124"	"4125"	"4126"	"4127"	"4128"	"4129"	"4130"
##	[4131]	"4131"	"4132"	"4133"	"4134"	"4135"	"4136"	"4137"	"4138"	"4139"	"4140"
##	[4141]	"4141"	"4142"	"4143"	"4144"	"4145"	"4146"	"4147"	"4148"	"4149"	"4150"
##	[4151]	"4151"	"4152"	"4153"	"4154"	"4155"	"4156"	"4157"	"4158"	"4159"	"4160"
##	[4161]	"4161"	"4162"	"4163"	"4164"	"4165"	"4166"	"4167"	"4168"	"4169"	"4170"
##	[4171]	"4171"	"4172"	"4173"	"4174"	"4175"	"4176"	"4177"	"4178"	"4179"	"4180"
##	[4181]	"4181"	"4182"	"4183"	"4184"	"4185"	"4186"	"4187"	"4188"	"4189"	"4190"
##	[4191]	"4191"	"4192"	"4193"	"4194"	"4195"	"4196"	"4197"	"4198"	"4199"	"4200"
##	[4201]	"4201"	"4202"	"4203"	"4204"	"4205"	"4206"	"4207"	"4208"	"4209"	"4210"
##	[4211]	"4211"	"4212"	"4213"	"4214"	"4215"	"4216"	"4217"	"4218"	"4219"	"4220"
##	[4221]	"4221"	"4222"	"4223"	"4224"	"4225"	"4226"	"4227"	"4228"	"4229"	"4230"
##	[4231]	"4231"	"4232"	"4233"	"4234"	"4235"	"4236"	"4237"	"4238"	"4239"	"4240"
##	[4241]	"4241"	"4242"	"4243"	"4244"	"4245"	"4246"	"4247"	"4248"	"4249"	"4250"
##	[4251]	"4251"	"4252"	"4253"	"4254"	"4255"	"4256"	"4257"	"4258"	"4259"	"4260"
##	[4261]	"4261"	"4262"	"4263"	"4264"	"4265"	"4266"	"4267"	"4268"	"4269"	"4270"
##	[4271]	"4271"	"4272"	"4273"	"4274"	"4275"	"4276"	"4277"	"4278"	"4279"	"4280"
##	[4281]	"4281"	"4282"	"4283"	"4284"	"4285"	"4286"	"4287"	"4288"	"4289"	"4290"
##	[4291]	"4291"	"4292"	"4293"	"4294"	"4295"	"4296"	"4297"	"4298"	"4299"	"4300"
##	[4301]	"4301"	"4302"	"4303"	"4304"	"4305"	"4306"	"4307"	"4308"	"4309"	"4310"
##	[4311]	"4311"	"4312"	"4313"	"4314"	"4315"	"4316"	"4317"	"4318"	"4319"	"4320"
##	[4321]	"4321"	"4322"	"4323"	"4324"	"4325"	"4326"	"4327"	"4328"	"4329"	"4330"
##	[4331]	"4331"	"4332"	"4333"	"4334"	"4335"	"4336"	"4337"	"4338"	"4339"	"4340"
##	[4341]	"4341"	"4342"	"4343"	"4344"	"4345"	"4346"	"4347"	"4348"	"4349"	"4350"
##	[4351]	"4351"	"4352"	"4353"	"4354"	"4355"	"4356"	"4357"	"4358"	"4359"	"4360"
##	[4361]	"4361"	"4362"	"4363"	"4364"	"4365"	"4366"	"4367"	"4368"	"4369"	"4370"
##	[4371]	"4371"	"4372"	"4373"	"4374"	"4375"	"4376"	"4377"	"4378"	"4379"	"4380"
##	[4381]	"4381"	"4382"	"4383"	"4384"	"4385"	"4386"	"4387"	"4388"	"4389"	"4390"
##	[4391]	"4391"	"4392"	"4393"	"4394"	"4395"	"4396"	"4397"	"4398"	"4399"	"4400"
##	[4401]	"4401"	"4402"	"4403"	"4404"	"4405"	"4406"	"4407"	"4408"	"4409"	"4410"
##	[4411]	"4411"	"4412"	"4413"	"4414"	"4415"	"4416"	"4417"	"4418"	"4419"	"4420"
##	[4421]	"4421"	"4422"	"4423"	"4424"	"4425"	"4426"	"4427"	"4428"	"4429"	"4430"
##	[4431]	"4431"	"4432"	"4433"	"4434"	"4435"					

```
## [4601] "4601" "4602" "4603" "4604" "4605" "4606" "4607" "4608" "4609" "4610"
## [4611] "4611" "4612" "4613" "4614" "4615" "4616" "4617" "4618" "4619" "4620"
## [4621] "4621" "4622" "4623"
```

```
ncol(Neonics)
```

```
## [1] 30
```

```
nrow(Neonics)
```

```
## [1] 4623
```

```
str(Neonics)
```

```
## 'data.frame': 4623 obs. of 30 variables:
## $ CAS.Number : int 58842209 58842209 58842209 58842209 58842209 58842209 58842209 58842209 58842209 58842209
## $ Chemical.Name : Factor w/ 9 levels "(1E)-N-[(6-Chloro-3-pyridinyl)methyl]-N-ethyl" ...
## $ Chemical.Grade : Factor w/ 9 levels "Analytical grade",...: 9 9 9 9 9 9 9 9 9 9
## $ Chemical.Analysis.Method : Factor w/ 5 levels "Measured","Not coded",...: 4 4 4 4 4 4 4 4 4 4
## $ Chemical.Purity : Factor w/ 80 levels ">=98",">=99.0",...: 69 69 50 50 50 50 50 50 50 50
## $ Species.Scientific.Name : Factor w/ 398 levels "Acalolepta vastator",...: 69 69 248 248 248 248 248 248 248 248
## $ Species.Common.Name : Factor w/ 303 levels "Alfalfa Leafcutter Bee",...: 74 74 142 142 142 142 142 142 142 142
## $ Species.Group : Factor w/ 4 levels "Insects/Spiders",...: 1 1 1 1 1 1 1 1 1 1
## $ Organism.Lifestage : Factor w/ 20 levels "Adult","Cocoon",...: 1 1 19 19 19 19 19 19 19 19
## $ Organism.Age : Factor w/ 39 levels "<=24","<=48",...: 39 39 39 39 39 39 36 39 36 39
## $ Organism.Age.Units : Factor w/ 11 levels "Day(s)","Days post-emergence",...: 9 9 4 4 4 4 4 4 4 4
## $ Exposure.Type : Factor w/ 24 levels "Choice","Dermal",...: 23 23 11 11 11 11 11 11 11 11
## $ Media.Type : Factor w/ 10 levels "Agar","Artificial soil",...: 7 7 3 3 3 3 3 3 3 3
## $ Test.Location : Factor w/ 4 levels "Field artificial",...: 4 4 4 4 4 4 4 4 4 4
## $ Number.of.Doses : Factor w/ 30 levels "' 4-5',' 4-7',...: 30 30 18 18 18 18 18 18 18 18
## $ Conc.1.Type..Author. : Factor w/ 3 levels "Active ingredient",...: 1 1 1 1 1 1 1 1 1 1
## $ Conc.1..Author. : Factor w/ 1006 levels "<0.0004","<0.025",...: 639 510 813 622 441 441 441 441 441 441
## $ Conc.1.Units..Author. : Factor w/ 148 levels "%","% v/v","% w/v",...: 132 132 91 91 91 91 91 91 91 91
## $ Effect : Factor w/ 19 levels "Accumulation",...: 16 16 16 16 16 16 16 16 16 16
## $ Effect.Measurement : Factor w/ 155 levels "Abundance","Accuracy of learned task, per",...: 15 15 8 8 8 8 8 8 8 8
## $ Endpoint : Factor w/ 28 levels "EC10","EC50",...: 15 15 8 8 8 8 8 8 8 8
## $ Response.Site : Factor w/ 19 levels "Abdomen","Brain",...: 14 14 14 14 14 14 14 14 14 14
## $ Observed.Duration..Days. : Factor w/ 361 levels "<.0002","<.0021",...: 145 145 145 145 145 145 145 145 145 145
## $ Observed.Duration.Units..Days. : Factor w/ 17 levels "Day(s)","Day(s) post-emergence",...: 1 1 1 1 1 1 1 1 1 1
## $ Author : Factor w/ 433 levels "Abbott,V.A., J.L. Nadeau, H.A. Higo, and M",...: 1 1 1 1 1 1 1 1 1 1
## $ Reference.Number : int 107388 107388 103312 103312 103312 103312 103312 103312 103312 103312
## $ Title : Factor w/ 458 levels "A Common Pesticide Decreases Foraging Suc",...: 1 1 1 1 1 1 1 1 1 1
## $ Source : Factor w/ 456 levels "Acta Hortic.1094:451-456",...: 295 295 296 296 296 296 296 296 296 296
## $ Publication.Year : int 1982 1982 1986 1986 1986 1986 1986 1986 1986 1986
## $ Summary.of.Additional.Parameters: Factor w/ 943 levels "Purity: \xca NC - NC | Organism Age: \xca",...: 1 1 1 1 1 1 1 1 1 1
```

```
#there are 30 columns or variables and 4623 rows or observations
```

- Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
summary(Neonics$Effect)
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry
##           12           102           360           11
##      Cell(s)      Development      Enzyme(s) Feeding behavior
##           9           136           62           255
##      Genetics      Growth      Histology      Hormone(s)
##          82           38           5           1
##      Immunological      Intoxication      Morphology      Mortality
##          16           12           22           1493
##      Physiology      Population      Reproduction
##           7           1803           197
```

```
# specifically looks at the column by using a $
```

Answer: The top 3 most common are population, mortality, and Behavior. These effects could definitely be of interest when considering how many of these insects are present at any given time in an agricultural place. Also, understanding the insect's behavior is critical in diagnosing how they interact with pollinators and crops. Finally, mortality is specifically interesting of how many of these insects are dying at any given time (unless its doing mortality of other entities dying) whether that's a result of pesticides or other methods.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name)
```

```
##      Honey Bee      Parasitic Wasp
##           667           285
##      Buff Tailed Bumblebee      Carniolan Honey Bee
##           183           152
##      Bumble Bee      Italian Honeybee
##          140           113
##      Japanese Beetle      Asian Lady Beetle
##           94           76
##      Euonymus Scale      Wireworm
##           75           69
##      European Dark Bee      Minute Pirate Bug
##           66           62
##      Asian Citrus Psyllid      Parastic Wasp
##           60           58
##      Colorado Potato Beetle      Parasitoid Wasp
##           57           51
##      Erythrina Gall Wasp      Beetle Order
##           49           47
##      Snout Beetle Family, Weevil      Sevenspotted Lady Beetle
##           47           46
##      True Bug Order      Buff-tailed Bumblebee
##           45           39
```

##	Aphid Family	Cabbage Looper
##	38	38
##	Sweetpotato Whitefly	Braconid Wasp
##	37	33
##	Cotton Aphid	Predatory Mite
##	33	33
##	Ladybird Beetle Family	Parasitoid
##	30	30
##	Scarab Beetle	Spring Tiphia
##	29	29
##	Thrip Order	Ground Beetle Family
##	29	27
##	Rove Beetle Family	Tobacco Aphid
##	27	27
##	Chalcid Wasp	Convergent Lady Beetle
##	25	25
##	Stingless Bee	Spider/Mite Class
##	25	24
##	Tobacco Flea Beetle	Citrus Leafminer
##	24	23
##	Ladybird Beetle	Mason Bee
##	23	22
##	Mosquito	Argentine Ant
##	22	21
##	Beetle	Flatheaded Appletree Borer
##	21	20
##	Horned Oak Gall Wasp	Leaf Beetle Family
##	20	20
##	Potato Leafhopper	Tooth-necked Fungus Beetle
##	20	20
##	Codling Moth	Black-spotted Lady Beetle
##	19	18
##	Calico Scale	Fairyfly Parasitoid
##	18	18
##	Lady Beetle	Minute Parasitic Wasps
##	18	18
##	Mirid Bug	Mulberry Pyralid
##	18	18
##	Silkworm	Vedalia Beetle
##	18	18
##	Araneoid Spider Order	Bee Order
##	17	17
##	Egg Parasitoid	Insect Class
##	17	17
##	Moth And Butterfly Order	Oystershell Scale Parasitoid
##	17	17
##	Hemlock Woolly Adelgid Lady Beetle	Hemlock Woolly Adelgid
##	16	16
##	Mite	Onion Thrip
##	16	16
##	Western Flower Thrips	Corn Earworm
##	15	14
##	Green Peach Aphid	House Fly
##	14	14

```
##              Ox Beetle              Red Scale Parasite
##              14              14
##      Spined Soldier Bug      Armoured Scale Family
##              14              13
##      Diamondback Moth              Eulophid Wasp
##              13              13
##      Monarch Butterfly      Predatory Bug
##              13              13
##      Yellow Fever Mosquito      Braconid Parasitoid
##              13              12
##      Common Thrip      Eastern Subterranean Termite
##              12              12
##      Jassid              Mite Order
##              12              12
##      Pea Aphid              Pond Wolf Spider
##              12              12
##      Spotless Ladybird Beetle      Glasshouse Potato Wasp
##              11              10
##      Lacewing      Southern House Mosquito
##              10              10
##      Two Spotted Lady Beetle      Ant Family
##              10              9
##      Apple Maggot      (Other)
##              9              670
```

```
sort(Neonics$Species.Common.Name, decreasing = TRUE)[1:6]
```

```
## [1] Yellow-faced Bumblebee Yellow-faced Bumblebee Yellow-faced Bumblebee
## [4] Yellow-faced Bumblebee Yellow Mealworm Beetle Yellow Fever Mosquito
## 303 Levels: Alfalfa Leafcutter Bee Alfalfa Plant Bug ... Yellow-faced Bumblebee
```

#looked back at some old assignments to try and simplify this

Answer: The six most common are: honey bee, parasitic wasp, buff tailed bumblebee, carniolan honey bee, bumble bee, and Italian Honeybee. Most of these are pollinators except for the parasitic wasp which mostly kills pests but all largely have value for agriculture. These pollinators are extremely vulnerable to pesticides and chemicals which is right at the heart of this study. It would also be interesting, given the presence of the wasp, that if they were to deliberately target pollinators, other species included in this study otherwise could also be at risk.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics)
```

```
## [1] "data.frame"
```

```
Neonics_df <- (Neonics)
Neonics_df <- as.data.frame(Neonics)
```

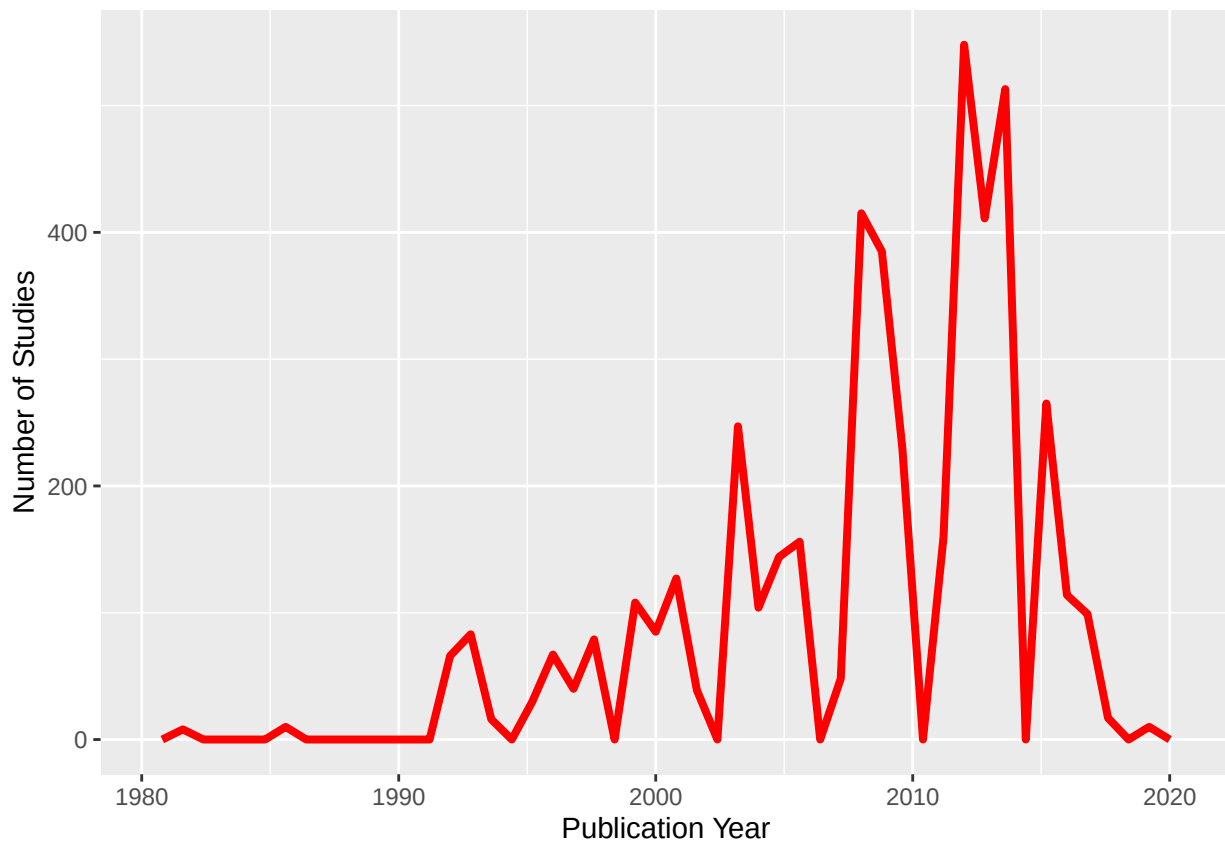
Answer: The class is a character class or text (strings) and it appears the reason it isn't numeric is due to values present that is not entirely numbers.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
library(ggplot2)
ggplot(Neonics, aes(x = Publication.Year)) +
  geom_freqpoly(binwidth = 0.8, color = "red", size = 1.4) +
  labs(
    x = "Publication Year",
    y = "Number of Studies"
  )
```

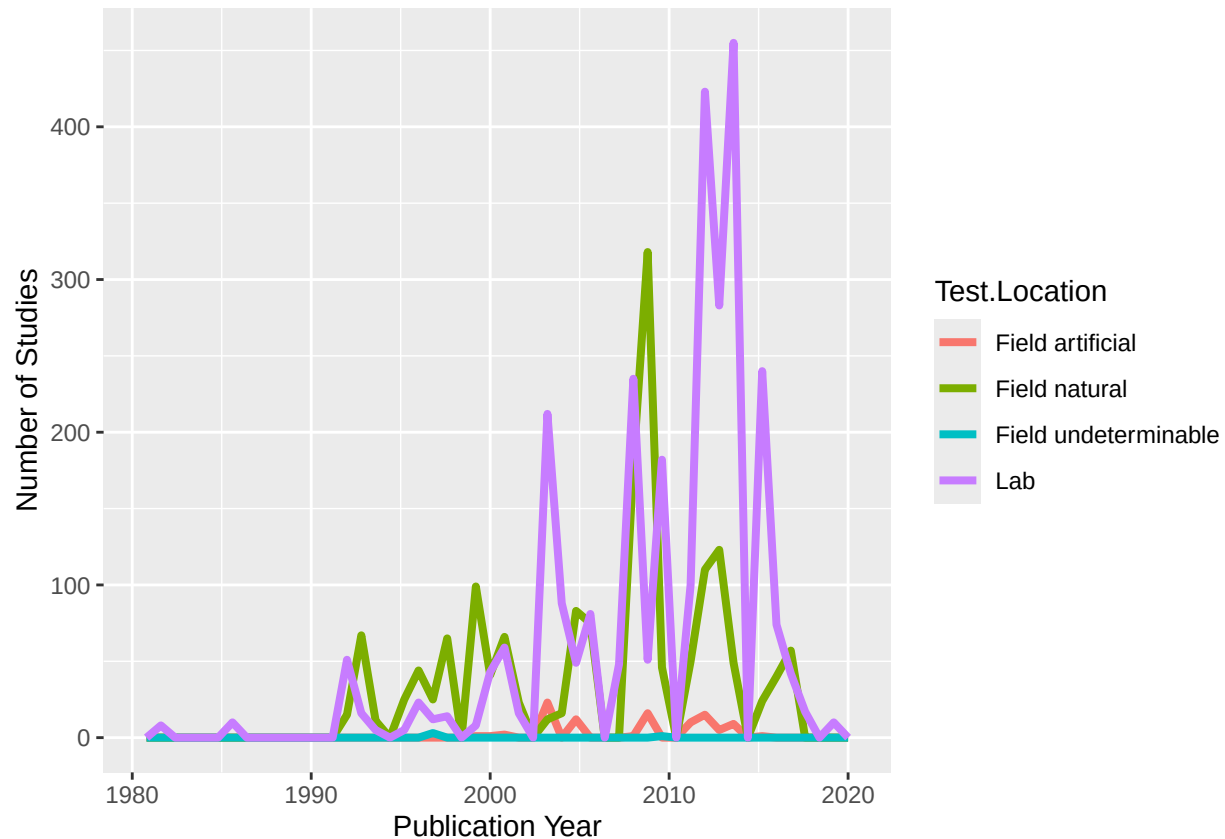
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



```
# played around with this one, liked the color red
view(Neonics)
```

10. Reproduce the same graph but now add a color aesthetic so that different `Test.Location` are displayed as different colors.

```
ggplot(Neonics, aes(x = Publication.Year, color = Test.Location)) +
  geom_freqpoly(binwidth = 0.8, size = 1.4) +
  labs(
    x = "Publication Year",
    y = "Number of Studies", Color = "Test Location"
  )
```



used largely same code, just moved around where the color was, feeling decently okay

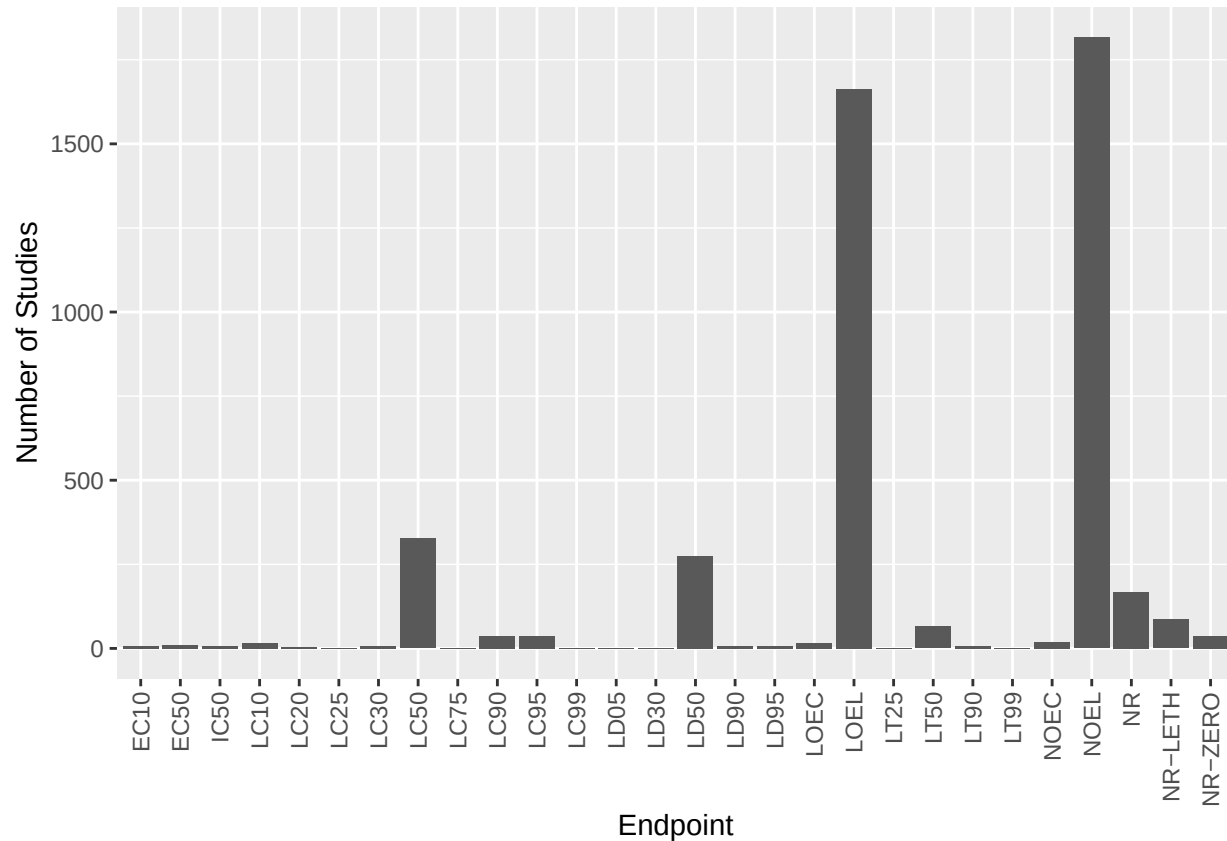
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: Laboratory studies were a major bulk of the tests. Field natural studies are the second most common and while in the early 90s they were the most popular study type, that changed after 2000. Field artificial studies are the most rare and largely were not used much except a point around 2002-2003 where they occurred more than the field natural studies. Finally, field undeterminable is a non-factor.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar() +
  labs(x = "Endpoint", y = "Number of Studies") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



How do you make the background of the graph white? I don't like the background at all.

Answer: The two most common end points were NOEL and LOEL. NOEL is defined as no observed effect level. LOEL is defined as low observed effect level. ## Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
Litter <- read_csv("Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv")
```

```
## Rows: 188 Columns: 19
## -- Column specification -----
## Delimiter: ","
## chr  (12): uid, namedLocation, domainID, siteID, plotID, trapID, fieldSample...
## dbl  (1): dryMass
## lgl  (1): remarks
## dtm  (2): ovenStartDate, ovenEndDate
```

```
## date (3): weighDate, setDate, collectDate
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
# yes it is a date, or it at least tells me its a "Date" in Console
Litter$collectDate <- as.Date(Litter$collectDate, format = "%Y-%m-%d")
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
# rechecked it to be safe
unique(Litter$collectDate[format(Litter$collectDate, "%Y-%m") == "2018-08"])
```

```
## [1] "2018-08-02" "2018-08-30"
```

```
# there are two applicable dates, August 2 and August 30
```

13. Using the unique function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from unique different from that obtained from summary?

```
unique(Litter$plotID)
```

```
## [1] "NIWO_061" "NIWO_064" "NIWO_067" "NIWO_040" "NIWO_041" "NIWO_063"
## [7] "NIWO_047" "NIWO_051" "NIWO_058" "NIWO_046" "NIWO_062" "NIWO_057"
```

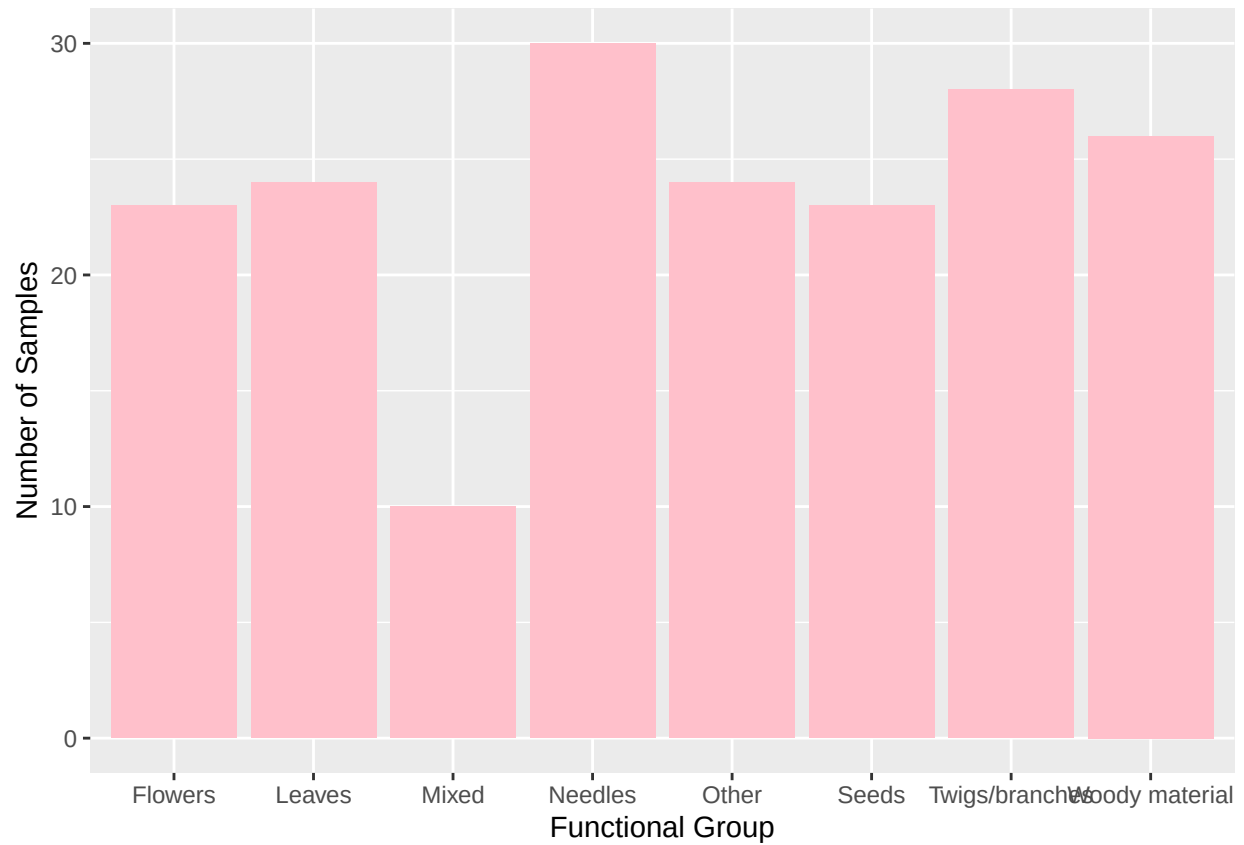
```
# "NIWO_061" "NIWO_064" "NIWO_067" "NIWO_040" "NIWO_041" "NIWO_063" "NIWO_047" "NIWO_051" "NIWO_058" "NIWO_057"
summary(Litter$plotID)
```

```
##      Length      Class      Mode
##      188 character character
```

```
#swappage of code
```

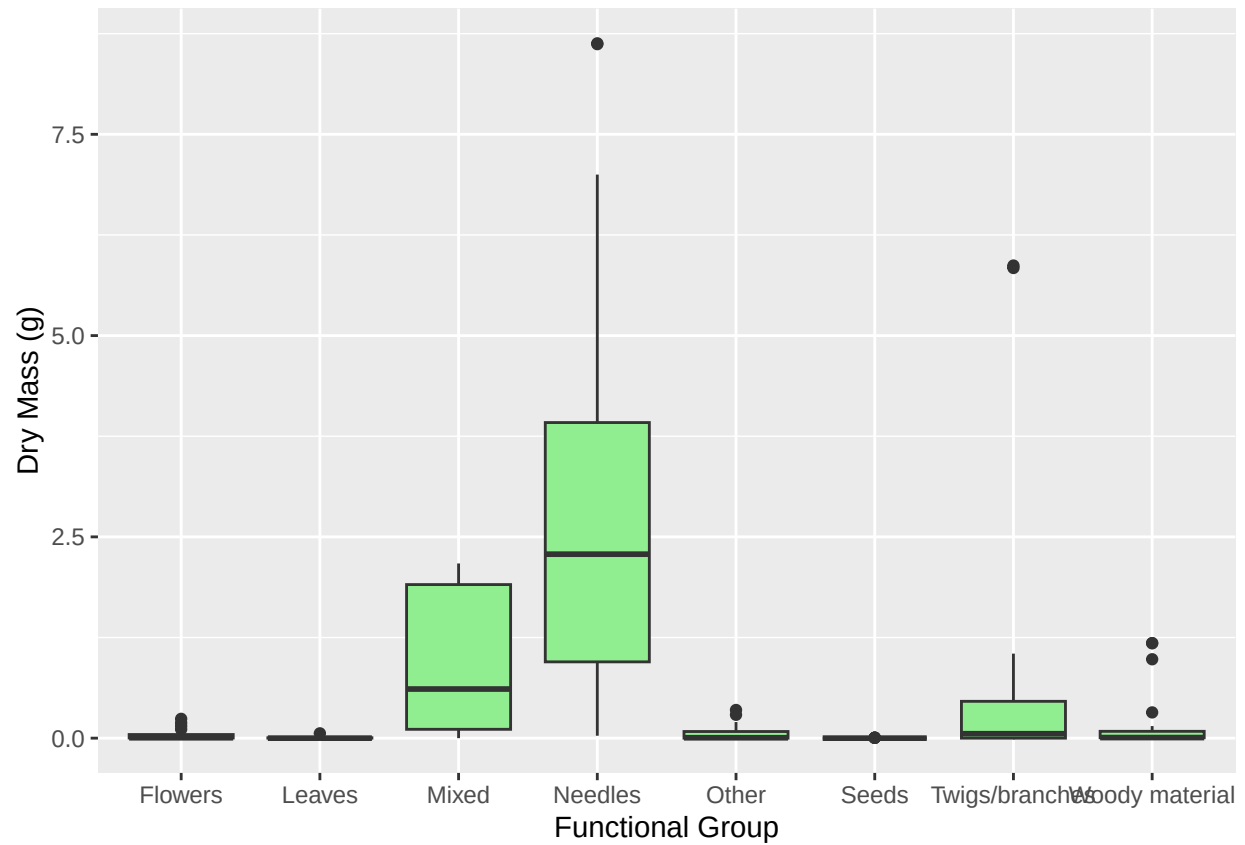
Answer: unique gives distinct values, summary is an overview so not as helpful 14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
library(ggplot2)
ggplot(Litter, aes(x = functionalGroup)) +
  geom_bar(fill = "pink") +
  labs(x = "Functional Group", y = "Number of Samples")
```

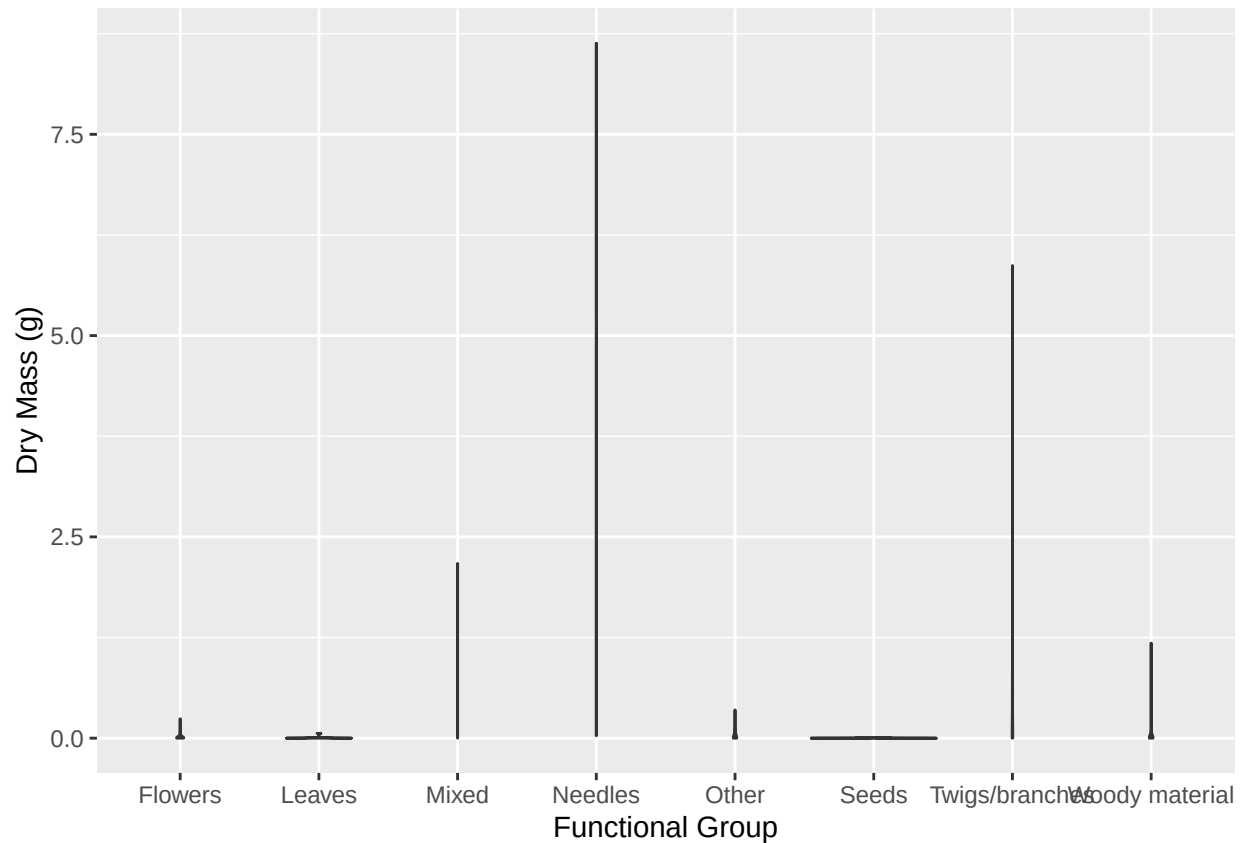


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +  
  geom_boxplot(fill = "lightgreen") +  
  labs(x = "Functional Group", y = "Dry Mass (g)")
```



```
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
  geom_violin(fill = "lightblue") +
  labs(x = "Functional Group", y = "Dry Mass (g)")
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: It clearly shows results that are easy to compare across the board. The violin plot is harder to read and is less clear to understand. Put it simply, I can read the box plot, the violin plot confuses me.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Based on the data, it appears needles, mixed, and twigs/branches tend to have the highest biomass. Effectively woody litter material gets the most biomass.